Personality-Aware Digital Coaching Through Adaptive Dialogue: Design, Implementation, and Pilot Evaluation with Swiss Caregivers

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Abstract

Digital coaching systems personalize through mood tracking or conversation memory, but neither mechanism captures the stable personality traits that determine how users respond to different coaching styles. This paper presents CareLoop, a personality-aware coaching architecture that adds this missing layer: continuous Big Five (OCEAN) trait inference from multi-turn dialogue, stabilized via exponential moving average (EMA) smoothing, and translated into behavioural directives that modulate emotional tone, relevance, and responsiveness. The system operates across four coaching modes—emotional support, practical education, policy navigation, and mixed—with hybrid vector and full-text retrieval and mandatory grounding verification. Three enforced pipeline invariants ensure that personality detection runs on every turn, evidence retrieval is conditional on factual need, and policy content is always citation-grounded. A five-layer hybrid intent router combines LLM-based classification with confidence-gated heuristic fallback and safety overrides. Automated pillar test matrices (130+ cases), latency and load benchmarks, and audit-trail validation provide pilot evidence of system reliability, routing accuracy, and grounding safety. Full-corpus validation against the PERSONAGE dataset—320 human-annotated restaurant-recommendation utterances with ground-truth Big Five ratings spanning all personality conditions—yields statistically significant Pearson correlations for all five traits: $r = 0.39$ (Openness), 0.42 (Conscientiousness), 0.38 (Extraversion), 0.59 (Agreeableness), and 0.50 (Neuroticism), with a mean $r = 0.46$ (all $p < 10^{-12}$). Implemented with Next.js, N8N orchestration, PostgreSQL/pgvector, and Meta Llama 3.3 70B via NVIDIA API, CareLoop demonstrates that personality-aware coaching modulation and factual grounding can coexist in a single auditable pipeline. Swiss informal caregivers serve as the pilot use case; the modular design is intended for domain transfer.

Keywords: personalized digital coaching, personality-aware dialogue, large language models, retrieval-augmented generation, Swiss caregivers, adaptive human-computer interaction

Contents

Abstract 1

1 Introduction 1

2 Related Work 4
- 2.1 Digital Coaching Systems and Personalisation Mechanisms . . . 4
- 2.2 Retrieval-Augmented Generation and Grounding . . . 5
- 2.3 Digital Support for Informal Caregivers . . . 6

3 System Design 8
- 3.1 Architecture and Pipeline Invariants . . . 8
- 3.2 Personality Detection and Stabilisation . . . 9
- 3.3 Hybrid Intent Router . . . 11
- 3.4 Retrieval and Grounding Verification . . . 11
- 3.5 Infrastructure, Audit, and Reproducibility . . . 12

4 Evaluation 13
- 4.1 Study Design and Validation Streams . . . 13
- 4.2 Baselines . . . 13
- 4.3 Measures . . . 13
- 4.4 Procedure . . . 13
- 4.5 Ethics . . . 13

5 Results 14
- 5.1 Personality Detection and Stabilisation (RQ1) . . . 14
- 5.2 Coaching Quality (RQ2) . . . 14
- 5.3 Grounding and Routing (RQ3 + RQ4) . . . 15
- 5.4 System Reliability (RQ5) . . . 15

6 Discussion 16
- 6.1 Interpretation and Contributions . . . 16
- 6.2 Reproducibility and Transferability . . . 16
- 6.3 Limitations and Future Work . . . 17

7 Conclusion 18

A Full Technology Stack 23

B Complete Trait-to-Directive Mapping 24

C Prompt Templates 25

D N8N Workflow Node Inventory 26

E Database Schema 27

F Benchmark Fixture Format 28

G Evaluation Rubric and Scoring Guidelines **29**

H Swiss Policy Corpus **30**

Declaration of Sole Authorship **31**

Chapter 1
Introduction

Effective coaching adapts to the individual—not merely in content but in style, pacing, and emotional register. Human coaches implicitly learn how a client processes information, tolerates uncertainty, handles stress, and prefers to receive feedback. This stable understanding of who the person is, rather than what the person said last session, is the basis of coaching effectiveness (Grant, 2003; Stober and Grant, 2006). As coaching moves into digital systems, the central design question is: what mechanisms can replicate this adaptation?

Current digital coaching applications personalize through three progressively deeper mechanisms. The first is *transient state tracking*: systems such as Woebot (Fitzpatrick et al., 2017) and Wysa (Inkster et al., 2018) detect the user's current mood and adapt the immediate response—selecting a breathing exercise for acute anxiety or a cognitive reframing prompt for rumination. This is reactive by design; it addresses the user's state at a single moment without modelling the stable traits that determine how they prefer to receive support. The second is *conversation memory*: LLM-based systems (ChatGPT Memory, mem0, LangChain Memory) store prior utterances and retrieve relevant facts across sessions. A system may recall that a caregiver mentioned a dementia diagnosis or asked about benefit eligibility. Memory enables continuity, but personalises *content*—what to reference—rather than *style*—how to communicate. It stores the user's history without modelling who they are. The third layer, largely absent from deployed systems, is *stable individual difference modelling*: inferring personality traits as enduring characteristics that should shape coaching behaviour across all interactions.

This gap is consequential. Coaching effectiveness is not personality-neutral: Big Five traits predict coaching outcomes, with Openness predicting benefit from exploratory approaches, Conscientiousness from structured goal-setting, and Neuroticism from emotional scaffolding (de Haan et al., 2013; Jones et al., 2016). A controlled field experiment confirmed that coachee personality and goal orientation moderate coaching-driven performance improvement (Smither et al., 2019). The same content delivered in different styles produces different outcomes depending on who receives it.

In informal caregiving specifically, personality shapes both how stress is experienced and which support strategies are effective. Neuroticism is a robust risk factor for elevated caregiver burden, depressive symptoms, and burnout: high-Neuroticism caregivers interpret demands as threats, exhibit poorer emotion regulation, and report greater subjective burden—a vulnerability that deepens over prolonged caregiving (Penley and Tomaka, 2002; Brodaty and Donkin, 2009). Conscientiousness functions as a protective factor through structured problem-solving, adherence to self-care routines, and lower perceived exhaustion. Agreeableness facilitates adaptive coping via social support-seeking and collaborative provider interaction. These trait-driven differences are operationally significant: a high-Neuroticism caregiver experiencing benefit-application anxiety requires emotional validation before procedural guidance; a high-Conscientiousness caregiver in the same situation prefers a structured checklist. Personality-coaching fit is not a stylistic preference—it determines whether an intervention is used at all.

The Big Five framework (OCEAN: Openness, Conscientiousness, Extraversion, Agreeableness, Neuroticism) provides the validated theoretical basis for this mechanism (Costa and McCrae, 1992; McCrae and John, 1992). Personality traits can be inferred from natural language with moderate to high accuracy (Yarkoni, 2010; Park et al., 2015), and personality-conditioned language generation improves perceived naturalness and satisfaction (Mairesse and Walker, 2010; Devdas, 2025). The Zurich Model of Social Motivation (Quirin et al., 2022, 2023) offers a theoretical bridge from OCEAN traits to coaching behaviour via three motivational systems—security, arousal, and affiliation—that predict preferred interaction patterns. The contribution of this work, however, is not personality detection but personality *modulation*: translating stabilised trait estimates into behavioural directives that continuously adapt coaching across three quality dimensions—emotional tone appropriateness, relevance and coherence, and responsiveness to emotional needs.

Modulation must coexist with factual accuracy, a requirement current coaching systems sidestep by avoiding domain-specific knowledge delivery. In healthcare policy contexts, a coaching response must simultaneously

validate a user's anxiety and provide citation-grounded procedural guidance. Rule-based CBT chatbots (Woebot, Wysa) deliver therapeutic exercises without domain retrieval. RAG-enhanced systems ground responses in evidence but apply no personality-aware modulation to how that evidence is presented. Ensuring that personality adaptation governs *how* content is delivered without altering *what* it says requires explicit architectural enforcement, not implicit reliance on the language model to maintain the distinction.

Swiss informal caregivers provide the concrete use case. Switzerland's Federal Statistical Office (FSO) reports life expectancy of 81.7 years for men and 85.7 for women (2022), with individuals aged 65+ already constituting nearly 20% of the population—a proportion projected to exceed 25% by 2040 as the "oldest-old" cohort (80+) is set to double by 2050 (Swiss Federal Statistical Office, 2023). The ratio of working-age adults to retirees has declined from 4:1 in 1990 to an anticipated 2:1 by 2040, creating a structural "care gap" in which informal caregivers currently provide approximately 80% of all long-term care. Approximately 700,000 individuals provide this unpaid care to family members with chronic illness, disability, or age-related needs (OECD, 2023; ZHAW, 2022). They face four intersecting stressor domains: physical demands from direct care tasks and disrupted sleep; emotional burden from witnessing decline and managing challenging behaviours (e.g., dementia-related agitation, which affects 40–80% of care recipients); financial strain from reduced employment and increased care costs; and social isolation from constrained participation and strained family relationships. Compounding this, Switzerland's Federal Health Insurance Act (KVG) covers only medically certified services and provides no direct compensation for informal caregiving; support is instead delegated to 26 cantons, creating a fragmented landscape where a caregiver in Zurich may access counselling and respite services while one in Appenzell receives virtually none. Swiss caregivers must navigate IV/Invalidenversicherung, AHV, Ergänzungsleistungen, cantonal Spitex services, and Hilflosenentschädigung in this heterogeneous policy environment. This domain simultaneously exercises all three personalisation layers—transient stress states, ongoing care memory, and stable personality traits—while demanding factually grounded policy guidance. It is thus an ideal proving ground for the present architecture. The system is designed for transfer to other coaching domains; Swiss caregivers are the use case, not the boundary.

The three-layer personalisation gap can be summarised as follows:

Table 1.1: Personalisation layer gap summary

Layer	Mechanism	Systems	Status
Transient state	Mood/sentiment detection	Woebot, Wysa	Addressed
Content recall	Conversation memory	ChatGPT Memory, mem0	Addressed
Stable individual differences	Personality trait modelling	None operational	Unaddressed
Coaching style adaptation	Trait-to-directive mapping	None operational	Unaddressed
Domain knowledge grounding	RAG with verification	Some medical systems	Not integrated with personality
Content/style separation	Architectural enforcement	None	Unaddressed

Mood tracking and memory are necessary—CareLoop employs both—but insufficient for personalised coaching. No existing system combines personality-trait modulation, domain knowledge retrieval, and mandatory grounding verification in a pipeline with formal invariants.

This study designs, implements, and evaluates CareLoop: a personality-aware coaching system that infers Big Five traits from multi-turn dialogue, stabilises them using EMA smoothing, and modulates coaching behaviour to improve emotional tone appropriateness, relevance and coherence, and emotional needs responsiveness. Four coaching modes—emotional support, practical education, policy navigation, and mixed—are governed by three pipeline invariants and classified through a hybrid LLM-and-heuristic intent router. The system is evaluated through automated pillar test matrices, latency and load benchmarks, and audit-trail validation, with Swiss informal caregivers as the pilot use case.

Table 1.2: Research questions

RQ	Focus
RQ1	Continuous OCEAN inference with EMA: valid, temporally stable trait estimates?
RQ2	Personality modulation improves emotional tone, relevance, emotional-needs responsiveness vs. non-adaptive baselines?
RQ3	RAG-enhanced policy navigation: full citation coverage, no critical grounding errors?
RQ4	Hybrid LLM router correctly classifies intent across four coaching modes with acceptable latency?
RQ5	System achieves acceptable reliability under load with full audit coverage?

Chapter 2
Related Work

2.1 Digital Coaching Systems and Personalisation Mechanisms

Digital coaching has emerged as a scalable intervention format, with the AI-driven mental health app market reaching $1.45 billion in 2024 (Grand View Research, 2024). Three architectural models dominate current deployments, each with characteristic personalisation capabilities and limitations.

Rule-based systems such as Woebot (Fitzpatrick et al., 2017) deliver CBT through pre-programmed decision trees, with mood-adaptive branching selecting exercise modules based on keyword-classified emotional state. Woebot's RCT demonstrated significant depression symptom reductions versus an information-only control (Fitzpatrick et al., 2017), validating the therapeutic value of structured digital coaching. However, two users with identical mood states but different personality profiles receive identical content and delivery style.

Hybrid NLP systems such as Wysa (Inkster et al., 2018) combine rule-based therapeutic modules with free-text NLP routing, crisis keyword detection, and working-alliance development through conversational rapport-building (Beatty et al., 2022). Personalisation is richer but still operates at the level of transient state and conversational context—the system adapts to what the user says and feels now, not to who they are across sessions.

LLM-based memory systems store user statements as vector embeddings for cross-session retrieval (Li et al., 2024). Comprehensive surveys identify three technical levels: prompting with personalised context, fine-tuning with personalised adapters, and alignment with personalised preferences (Chen et al., 2025; Wang et al., 2026). These systems enable impressive conversational continuity but personalise *content recall* rather than *coaching style*. A memory system knows what the user said; it does not know who the user is.

The personality gap. Across all three models, no deployed coaching system models the stable individual differences that predict how users respond to different coaching styles. Big Five traits moderate coaching outcomes—Openness predicts benefit from exploratory approaches, Conscientiousness from structured goal-setting, Neuroticism from emotional scaffolding (de Haan et al., 2013; Jones et al., 2016; Smither et al., 2019)—yet no digital system has operationalised this evidence. The cost of this gap is not merely suboptimal outcomes but measurably higher attrition: personality-blind systems produce emotionally incongruent responses that high-Neuroticism users experience as invalidating, amplifying rather than buffering the negative affect that originally drove them to seek support, and making disengagement more likely than continued engagement with the non-adaptive tool. Digital health attrition research confirms this mechanism: interventions incorporating emotional validation and personalised encouragement sustain engagement for users in high-stress states, whilst those delivering generic content to emotionally activated users accelerate dropout. Critically, modern personality science treats traits not as fixed labels applied once but as *distributions of states* that activate differentially across situational contexts: extraversion increases in social settings, neuroticism spikes under stress, and conscientiousness rises during task-oriented contexts (Fleeson, 2001). Experience sampling studies confirm that momentary personality states predict user outcomes—including engagement, well-being, and intervention response—independently of static trait scores, and that effective personalisation therefore requires per-turn dynamic inference rather than a single baseline assessment. No existing digital coaching system performs continuous within-session personality inference.

Personality-aware dialogue. The technical foundations for bridging this gap exist. Yarkoni (2010) demonstrated systematic associations between word-use patterns and personality dimensions; Park et al. (2015) achieved personality prediction from social media text. In dialogue generation, Mairesse and Walker (2010) showed that personality-conditioned language improves perceived naturalness; Hu et al. (2018) extended this to joint modelling of linguistic style and personality adaptation in NLG. Subsequent work explored persona-based dialogue models (Li et al., 2016; Zheng et al., 2019). Devdas (2025) advanced continuous Big Five detection with motivational regulation, demonstrating at least 20% improvement in emotional support quality. The Zurich Model of Social Motivation (Quirin et al., 2022, 2023) provides a theoretical bridge from OCEAN to coaching be-

haviour through three motivational systems—security (Neuroticism), arousal (Openness, Extraversion), and affiliation (Agreeableness)—with Conscientiousness as a structural modifier.

Two deployment problems remain unsolved. First, *temporal stability*: persona-aware generation conditions on static profiles (Li et al., 2016; Majumder et al., 2020) without modelling how personality estimates should evolve and stabilise across turns. The field confirms that "traditional approaches often rely on static personality profiles," with dynamic representation identified as an open problem (Zheng et al., 2019). Recent systems address *within-session* personality dynamism—COSPLAY (Xu et al., 2022) represents self- and partner personas as evolving concept sets; the Dual Enhanced Network (Zhu et al., 2024) jointly encodes long-term traits and short-term states for detection; Wen et al. (2024) formalises personality as a transition weight in per-turn emotion generation—but none address *cross-session stabilisation* of noisy per-turn detections into persistent persona profiles. A further open problem is identity drift: LLMs conditioned on personality profiles exhibit deteriorating Rank-Order stability under prolonged interaction, with core traits increasingly contradicted as conversations lengthen (Kovač et al., 2024). All dynamic approaches require model training on annotated dialogue corpora unavailable for Swiss caregiver scenarios. Second, *classification to behaviour*: adaptive dialogue management has approached this through reinforcement learning, meta-learning, and multi-task learning (den Hengst et al., 2019; Lee et al., 2021; Cordier et al., 2023). RL-based personalisation—including persona-consistent offline RL (Shea and Yu, 2023)—requires per-domain reward function design and thousands of training interactions before an initial policy is ready. Meta-learning approaches—such as personalising dialogue agents via MAML-style adaptation (Lin et al., 2019)—achieve few-shot personalisation from dialogue history but require a meta-training phase across multiple users, making them unsuitable for cold-start deployment with populations (Swiss caregivers) for whom no prior interaction corpus exists. Both families of methods also produce policies difficult to audit at the level of individual decisions. No prior system has combined continuous trait detection, cross-session temporal stabilisation, and formal directive-based modulation in an operational coaching pipeline without training data. The Zurich Model operationalisation in this work—mapping OCEAN traits to behavioural directives through the security, arousal, and affiliation systems—is absent from the prior dialogue literature.

2.2 Retrieval-Augmented Generation and Grounding

Retrieval-augmented generation (RAG) addresses the well-documented tendency of LLMs to generate plausible but factually unsupported content—hallucination (Ji et al., 2023). Lewis et al. (2020) introduced the RAG framework, demonstrating that conditioning generation on retrieved documents improves factual accuracy across knowledge-intensive tasks. Hybrid retrieval strategies—combining dense vector similarity and sparse lexical matching—achieve superior recall compared to either method alone (Karpukhin et al., 2020; Ma et al., 2023).

In health-adjacent and policy-guidance domains, grounding is a safety requirement rather than a quality improvement. Incorrect guidance about benefit eligibility or application procedures can cause material harm to vulnerable populations. Multi-stage grounding verification—extracting claims, checking evidence overlap, and flagging or replacing unsupported assertions—has emerged as a necessary complement to retrieval (Thoppilan et al., 2022; Gao et al., 2023; Singhal et al., 2023). A comprehensive survey of hallucination mitigation techniques identifies post-generation verification as the most reliable approach for deployment contexts where retrieval quality cannot be guaranteed (Tonmoy et al., 2024). Within post-generation verification, Chain-of-Verification (CoVe) establishes the principle that decomposing a response into independently verifiable claims—rather than assessing the whole response at once—prevents undetected hallucination through evidence masking, reducing unverified claim rates by up to 28% on open-domain factual tasks (Dhuliawala et al., 2023). Fine-grained claim-level detection further demonstrates that atomic sentence decomposition substantially outperforms response-level consistency scoring (Mishra et al., 2024).

Knowledge graph (KG) integration has been explored for enriching persona consistency—encoding persona attributes, interests, and belief structures to ensure character-coherent responses (Majumder et al., 2020; Luan et al., 2017). KG-based persona consistency is, however, distinct from factual policy grounding: the former ensures the system's persona remains internally coherent, whilst the latter enforces that specific policy claims trace to authoritative sources. No prior system has defined formal, enforced invariants that simultaneously require per-turn personality detection, conditional evidence retrieval, and citation-mandatory grounding. This combination is an architectural contribution distinct from either KG persona consistency or standard RAG deployment.

CareLoop integrates RAG with personality-aware coaching through explicit architectural separation: retrieval and grounding verification govern *what* the system says, whilst personality modulation governs *how* it says it.

Three enforced pipeline invariants maintain this separation. To our knowledge, this is the first system to treat personality modulation and factual grounding as simultaneously required but architecturally separated pipeline properties.

2.3 Digital Support for Informal Caregivers

Informal caregivers experience chronic stress, emotional exhaustion, and social isolation at rates significantly exceeding the general population (Schulz and Sherwood, 2008; Pinquart and Sörensen, 2003). Their stressor profile spans four intersecting domains: physical demands from direct care tasks and sleep disruption; emotional burden from witnessing decline and managing behavioural symptoms such as verbal agitation (present in 40–80% of dementia care recipients); financial strain from reduced employment and increased care costs; and social isolation from constrained participation and strained family relationships (Brodaty and Donkin, 2009). Digital interventions have shown promise—Mooney et al. (2024) reported a 38% burden reduction at 8 weeks—but are severely constrained by a personalisation gap that no existing tool has addressed.

Systematic reviews of digital caregiver interventions reveal a consistent pattern: high attrition rates, with many caregivers disengaging after only a few interactions, and limited efficacy in studies that do not account for individual differences. Research on digital health attrition identifies six primary driver categories—usability, perceived value, technical barriers, lack of personalisation, emotional and psychological factors, and contextual factors such as disease burden and demographics. For caregiver populations, the emotional/psychological and personalisation categories are dominant: caregivers contact digital tools precisely during acute stress episodes, and a generic system that delivers the same content regardless of the caregiver's current state provides a response that is emotionally incongruent—experienced not merely as irrelevant but as invalidating, which amplifies negative affect and actively accelerates disengagement rather than supporting it. Conversely, digital interventions incorporating emotional validation, positive reinforcement, and personalised encouragement sustain engagement for users in high-stress states.

Personality traits are significant moderators of both engagement and outcome. Neuroticism predicts higher attrition: high-Neuroticism individuals initially seek digital tools under acute stress but disengage rapidly when tools fail to provide emotionally congruent responses—the causal mechanism is that uniform delivery is interpreted as dismissal, amplifying rather than buffering emotional arousal. Conscientiousness predicts sustained engagement and adherence to structured features. Agreeableness predicts responsiveness to empathetic conversational agents. A single generic intervention design systematically fails the users who need it most: high-Neuroticism caregivers would benefit from emotionally calibrated support but abandon systems that deliver uniform, non-responsive responses. No existing caregiver tool integrates personality assessment with adaptive response style.

The Swiss context adds a domain-specific dimension. Switzerland's federalised healthcare system produces canton-specific variations in policy eligibility and service availability. Caregivers must navigate IV/Invalidenversicherung, AHV, Ergänzungsleistungen, cantonal Spitex services, and Hilflosenentschädigung—a fragmented landscape demanding both emotional resilience to engage with complexity and accurate factual guidance to act on it. Social robot interventions such as ElliQ (Broadbent et al., 2024) and evaluation frameworks such as FEEL (Zhang et al., 2024) improve scalability but lack personality adaptation and Swiss-specific knowledge grounding. CBT-based coaching apps address emotional wellbeing through therapeutic exercises but do not deliver grounded policy guidance.

The closest Swiss-specific prior work is the Caregiving Language Model (CaLM), which uses retrieval-augmented generation grounded in Swiss KVG, Disability Insurance (IV), Spitex association manuals, and Pro Senectute guidelines to provide reliable domain-specific guidance to family caregivers (Schardong and Custódio, 2021). CaLM demonstrates the feasibility and value of RAG-grounded, policy-compliant responses in this domain—but it provides no personality-aware adaptation and delivers the same response style to all caregivers regardless of trait profile. CareLoop builds on CaLM's knowledge-grounding principle whilst adding the personality detection and directive modulation layers that CaLM does not address.

A further design affordance specific to the Swiss population is the cultural norm of privacy and self-reliance. Swiss caregiving culture strongly discourages public disclosure of family difficulties and institutional help-seeking, which is perceived as a breach of familial loyalty. This creates a paradox: those who most need support are least likely to seek it through formal channels. Discrete, anonymous digital coaching provides a culturally congruent low-stigma entry point that does not require caregivers to identify themselves or their families to any institution, making conversational AI a particularly well-matched support modality for this population.

No existing caregiver support tool integrates transient state tracking, conversation memory, stable personality modelling, and citation-grounded domain knowledge in a single adaptive system. Comprehensive surveys of the personality-aware digital coaching field—covering hundreds of papers on personality representation, adaptive dialogue, and evaluation methodology—report no caregiver-facing implementations, no caregiver-specific datasets, and no systems designed for Swiss policy contexts. The present study addresses a domain that is both practically urgent and academically unexplored.

Chapter 3
System Design

3.1 Architecture and Pipeline Invariants

CareLoop is implemented as a three-tier architecture: a Next.js API route handles client requests and orchestrates pre- and post-processing; an N8N workflow engine executes the 9-stage coaching pipeline; and PostgreSQL with pgvector provides persistent storage for sessions, personality states, policy evidence, and audit records. Meta Llama 3.3 70B (70-billion-parameter instruction-tuned model), accessed via the NVIDIA API, serves as the language model for personality detection, response generation, and intent routing.

Each conversational turn proceeds through three phases. *API-side pre-processing* performs hybrid LLM intent routing, resolves session isolation and route keys, and conditionally executes vector and full-text retrieval to prefetch policy evidence. The *N8N workflow* then executes nine stages in sequence: input normalisation and context extraction; loading of prior personality states from the database; OCEAN trait detection with EMA smoothing; regulation through trait-to-directive mapping; policy retrieval merging prefetched evidence with workflow full-text search results; directive-conditioned response generation; grounding verification through claim extraction and evidence matching; quality verification and safety checking; and response formatting with persistence of session, turn, personality, and metrics data. *API-side post-processing* applies mode override and confidence propagation, performs optional LLM regeneration for heuristic-generated content, overlays citations, executes the grounding invariant check, and writes dual audit records.

Three pipeline invariants govern every turn regardless of coaching mode. These are enforced architectural rules, not guidelines:

Table 3.1: Pipeline invariants

Invariant	Rule	Enforcement
Universal personality	OCEAN detection and regulation run for every turn in every mode	N8N executes Detection → EMA → Regulation before mode-specific branching; API flags `ocean_detection_skipped` or `regulation_skipped` on violation
Conditional retrieval	Evidence retrieval runs only when factual content is needed	API-side gating function evaluates resolved mode, keyword signals, and policy opt-out phrases before initiating retrieval
Mandatory grounding	Policy and mixed responses must include citations	N8N grounding verifier checks claim-evidence overlap; API checks citation count post-response; zero citations triggers a grounding warning

The first invariant ensures every response is personality-informed regardless of coaching mode. The second prevents unnecessary retrieval latency on emotional or educational turns. The third guarantees no policy claim reaches the user without evidence support.

The four coaching modes define how the system balances personality modulation and factual grounding:

Table 3.2: Coaching modes

Mode	Personality modulation	Retrieval	Grounding
Emotional support	Full (tone, warmth, pacing)	Skipped	Not required
Practical education	Full (structure, detail, style)	Conditional	If citations present
Policy navigation	Presentation style only	Always	Mandatory
Mixed	Full modulation + presentation style	Always	Mandatory for policy segment

The mixed mode is architecturally significant: when a caregiver expresses emotional distress and asks a policy question in the same message, the system composes a response with both an emotionally supportive segment and a citation-grounded policy segment, applying personality modulation throughout whilst maintaining factual integrity in the policy portion.

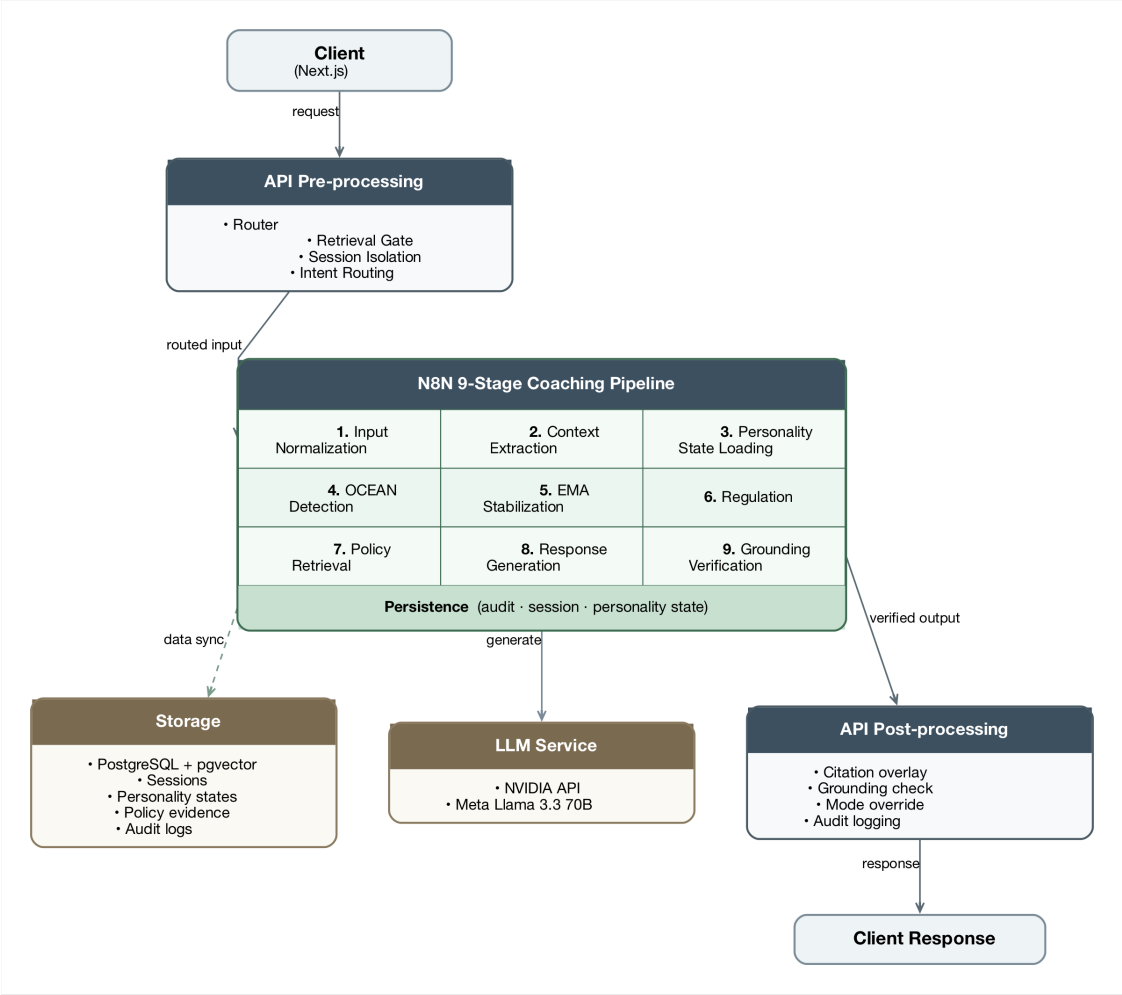

Figure 3.1: CareLoop three-tier architecture: Next.js API route, N8N 9-stage orchestration pipeline (detect → stabilise → regulate → retrieve → generate → verify → persist → audit), and PostgreSQL/pgvector storage layer. Personality detection and EMA stabilisation run on every turn (invariant 1); retrieval activates conditionally (invariant 2); all policy content is citation-grounded (invariant 3).

3.2 Personality Detection and Stabilisation

Personality detection, temporal stabilisation, and trait-to-directive modulation form a tightly coupled subsystem that runs on every turn, enforcing the universal personality invariant.

Detection. Each turn, the system infers current Big Five trait expression using Meta Llama 3.3 70B at temperature 0.1 via few-shot prompting. The prompt comprises three components: (1) a calibrated system message defining trait-specific linguistic markers for each OCEAN dimension; (2) seven domain-representative few-shot

examples spanning the full OCEAN range, each pairing a user utterance with the expected JSON trait scores; and (3) the user's current message. The model returns per-trait values in $[-1.0, +1.0]$ with per-trait confidence set to 0.8 for API-sourced detections. The few-shot calibration is critical: zero-shot prompting produced trait inversions (e.g., negative Conscientiousness correlation) because the model conflated topic sentiment with speaker personality style. A heuristic keyword fallback activates when the LLM is unavailable, providing approximate trait estimates at confidence 0.45.

EMA stabilisation. Raw per-turn estimates are noisy. The system addresses this through a hierarchical state estimation pipeline: LLM observations → session state estimation → long-term trait estimation → policy activation. This four-layer architecture operates across two time scales, mirroring experience sampling methodology (ESM/EMA) in personality science: within-person *personality states* fluctuate moment-to-moment, whilst between-person *personality traits* represent stable long-term baselines (Fleeson, 2001; Nesselroade, 1991).

Layer 1 — Observation. Each turn produces independent per-trait noisy observations with confidence scores: $(\hat{\mathbf{T}}^{(n)}, \mathbf{c}^{(n)}) = f_{\text{LLM}}(u^{(n)}, H^{(n-6:n-1)})$, $\hat{\mathbf{T}}^{(n)} \in [-1,1]^5$, where $u^{(n)}$ is the user message at turn n, H is the 6-turn dialogue history, and each trait dimension $i \in \{\text{O,C,E,A,N}\}$ has confidence $c_i^{(n)} \in [0,1]$.

Layer 2 — In-session state estimator. Per-trait confidence gating prevents low-quality observations from corrupting the session estimate. For each trait dimension i independently:

$$T_{\text{sess},i}^{(n)} = \begin{cases} \alpha_s \cdot \hat{T}_i^{(n)} + (1-\alpha_s) \cdot T_{\text{sess},i}^{(n-1)} & \text{if } c_i^{(n)} \geq \tau_c \\ T_{\text{sess},i}^{(n-1)} & \text{otherwise} \end{cases} \tag{3.1}$$

with $\alpha_s = 0.3$ and confidence gate $\tau_c = 0.4$. The session is initialised from the prior persona profile: $T_{\text{sess},i}^{(0)} = P_i^{(k-1)}$. Sensitivity analysis across $\alpha_s \in \{0.1, 0.2, 0.3, 0.4, 0.5\}$ shows convergence to within ± 0.05 of the underlying trait signal within 6–8 turns.

Layer 3 — Cross-session trait estimator. After each session k concludes: $P_i^{(k)} = \alpha_p \cdot T_{\text{sess},i}^{(\text{final})} + (1-\alpha_p) \cdot P_i^{(k-1)}$ with $\alpha_p = 0.2$. The effective memory horizon is $1/\alpha_p = 5$ sessions.

This two-level EMA structure is a fixed-gain approximation to a Kalman filter: when the latent trait evolves slowly, the Kalman gain converges toward a constant value, and the recursive update reduces to exponential moving average smoothing. The confidence gate τ_c is a computationally efficient hard approximation to continuous noise weighting.

Layer 4 — Directive activation. Stabilised session estimates trigger behavioural directives when trait magnitude exceeds $\theta_T = 0.15$: $\mathscr{D}^{(n)} = \{d_i : |T_{\text{sess},i}^{(n)}| > \theta_T\}$.

Table 3.3: Trait-to-directive mapping (summary)

Trait direction	Behavioural directive	Activation
High Neuroticism	Offer extra comfort; acknowledge anxieties	$T_{\text{sess},N} > 0.15$
Low Neuroticism	Reassure stability and confidence	$T_{\text{sess},N} < -0.15$
High Agreeableness	Show warmth, empathy, collaboration	$T_{\text{sess},A} > 0.15$
High Conscientiousness	Provide organised, structured guidance	$T_{\text{sess},C} > 0.15$
High Openness	Invite exploration and novelty	$T_{\text{sess},O} > 0.15$
High Extraversion	Energetic, sociable tone	$T_{\text{sess},E} > 0.15$

When no trait exceeds the activation threshold, default directives apply: warm, attentive, and supportive with open-ended questions. The directive mapping operationalises documented empirical patterns (e.g., high-Neuroticism caregivers interpret demands as threats; high-Conscientiousness employ structured problem-solving) (Carver et al., 1989; Lazarus and Folkman, 1984). CareLoop implements this through explicit trait-to-directive rules rather than learned discriminators, preserving interpretability and auditability (Asghar et al., 2020).

Figure 3.2: EMA trait stabilisation: raw per-turn Neuroticism detections (volatile) versus EMA-smoothed in-session estimate and cross-session trait baseline across 8 turns. Confidence gating at turns with $c_i^{(n)} < \tau_c = 0.4$ preserves the prior estimate. The EMA estimate converges within 6 turns to a stable neighbourhood.

3.3 Hybrid Intent Router

Classifying the user's coaching need determines which pipeline branches activate—particularly whether evidence retrieval and mandatory grounding are triggered. Context-aware NLU systems that incorporate dialogue history consistently outperform single-turn classifiers (Gupta et al., 2019); however, large pretrained models are prone to overconfidence on out-of-domain inputs, motivating confidence gating as a rejection mechanism (Wang et al., 2024). CareLoop's router spans both task-oriented intents (policy navigation, practical education) and open-domain social intents (emotional support, mixed)—a combination absent from prior intent classification systems (Liu et al., 2024).

The system uses a five-layer hybrid router: (1) **Explicit requested mode**—frontend-specified modes used directly with confidence 1.0; (2) **LLM router**—Llama 3.3 70B at temperature 0 returns structured JSON (mode, confidence, retrieval flag, routing reason); results cached per message-context key for 5 minutes; conversation-history context (prior 3 turns) included; (3) **Confidence gate**—confidence below 0.5 discards the LLM decision and falls through to heuristics; (4) **Keyword heuristic fallback**—curated term lists (26 policy, 17 education, 15 emotional terms) with conversation-history inheritance for short follow-up messages; (5) **Hard safety override**—policy-related keywords force retrieval activation regardless of routing source. The N8N V2 workflow consumes the API routing decision directly. Every response carries routing metadata (classification source, routing reason, confidence score, LLM latency).

3.4 Retrieval and Grounding Verification

Retrieval. When the resolved mode requires evidence, the system executes hybrid retrieval. The primary path uses the NVIDIA embedding API (model `nv-embedqa-e5-v5`, 1024-dimensional vectors) for cosine similarity against pre-embedded policy chunks in PostgreSQL/pgvector, returning the top 5 results above a 0.30 similarity threshold. When vector search returns fewer than 2 results, PostgreSQL full-text search (`websearch_to_tsquery`) supplements and deduplicates. The policy corpus covers selected Swiss social insurance domains (IV eligibility, Zurich supplementary benefits, care subsidies, Hilflosenentschädigung). Personality modulation does not alter retrieved content—it governs only how that content is presented.

Grounding verification. A three-stage pipeline ensures no policy claim reaches the user without evidence support. *Stage 1:* The N8N Grounding Verifier node extracts policy claims via regex patterns and checks keyword overlap against retrieved evidence. A grounding ratio below 0.5 triggers full safe-fallback replacement; 0.5–0.8 produces a degraded response replacing ungrounded claims with "[information pending verification]"; ≥ 0.8 passes. *Stage 2:* The Verification and Refinement node scores overall adherence (session validity, question count, word count, banned claim phrases); scores below 0.5 trigger a full block. *Stage 3:* The API performs a post-response citation count check for policy and mixed mode responses; zero citations generates a `warn_no_citations` audit warning—a final backstop that no prior RAG system implements as a formal pipeline invariant.

3.5 Infrastructure, Audit, and Reproducibility

CareLoop is deployed via Docker Compose: a Next.js 14 frontend with Zustand state management; N8N self-hosted with task runners; PostgreSQL 16 with pgvector across 11 tables (sessions, turns, personality states, policy evidence, metrics, audit logs, users, feedback, memory embeddings, user personality profiles, policy chunks); and Meta Llama 3.3 70B via NVIDIA API. Inter-module contracts are validated using Zod schemas in a shared contracts package; user authentication uses HS256 JWT tokens in HTTP-only cookies.

Every turn produces a dual audit record—JSONL log and PostgreSQL `audit_log` row—containing: request ID, session ID, turn index, coaching mode, pipeline status, routing metadata, personality state (OCEAN scores only, no PII), retrieval IDs, citation count, verifier status, SHA-256 input hash, and turn latency. PII redaction precedes all logging.

Reproducibility is supported by: fixed model versions and parameters (Llama 3.3 70B, EMA $\alpha_s = 0.3$); versioned N8N workflow files (V-numbering convention); deterministic JSON contracts between all modules; containerised single-command deployment; and benchmark fixtures stored as versioned JSON enabling automated regression testing.

Chapter 4
Evaluation

4.1 Study Design and Validation Streams

This study follows a pilot system design and evaluation approach. Four validation streams provide converging evidence:

Table 4.1: Validation streams

Stream	Sample	What it validates
PERSONAGE benchmark	320 human-annotated utterances with ground-truth OCEAN (Mairesse and Walker, 2011)	External trait detection validity (Pearson r, MAE)
Automated pillar test matrices	130+ cases across 4 modes	Mode classification, citation presence, grounding safety, latency gates
Benchmark and load tests	10 expansion + 4 latency + 100 concurrent sessions	Performance gates, reliability under load
Expert pilot sessions	$n = 5$–8 domain experts (Spitex coordinators, geriatricians, home care nurses)	Usability, appropriateness, ecological validity
Simulated evaluation	$N \geq 250$ synthetic conversations across personality profiles and stress scenarios	Coaching quality compared against baselines

The PERSONAGE benchmark (Mairesse and Walker, 2011) provides external validation of trait detection accuracy against human-annotated ground truth. PERSONAGE contains restaurant-recommendation utterances systematically generated to express controlled Big Five personality profiles, with each utterance rated by human judges on all five OCEAN dimensions (1–7 Likert scale, normalised to $[-1, +1]$). After filtering for complete trait annotations, 320 utterances remain: 160 from predefined personality conditions and 160 from randomised parameter combinations. Real caregiver validation ($n = 20$–30) is fully specified as a future protocol, deferred due to recruitment feasibility constraints.

4.2 Baselines

Four conditions isolate the contribution of each personalisation layer:

Table 4.2: Experimental conditions and personalisation layers

Condition	Personalisation layers active	What it isolates
Personality-aware (full system)	Mood + memory + personality + RAG	Full personalised coaching
Generic non-adaptive	None (static responses)	Whether any personalisation matters
Memory-only	Memory + mood (no trait modelling)	Whether conversational continuity substitutes for personality modulation
Policy-only	RAG (no personality or memory)	Whether factual accuracy alone suffices

The memory-only versus full-system comparison is particularly informative: memory represents the current state-of-the-art in LLM personalisation, and any difference isolates the specific value of personality-trait modelling.

4.3 Measures

Table 4.3: Evaluation metrics, targets, and RQ mapping

Construct	Metric	Target	RQ
Trait validity (external)	Pearson r with PERSONAGE ground truth	≥ 0.40 (moderate); ≥ 0.75 (strong)	RQ1
Trait validity (internal)	Pearson r with synthetic profiles	≥ 0.75	RQ1
Trait stability	Variance within 6+ turns	< 0.05	RQ1
Confidence calibration	High-confidence predictions aligned	$\geq 85\%$	RQ1
Emotional tone appropriateness	LLM evaluator (CoT, temp 0) + human expert	$\geq 4.0/5.0$	RQ2
Relevance & coherence	LLM evaluator (CoT, temp 0) + human expert	$\geq 4.0/5.0$	RQ2
Emotional needs responsiveness	Improvement over baseline	$\geq 20\%$	RQ2
Citation coverage	Policy claims with sources	100%	RQ3
Grounding errors	Critical hallucinations	0	RQ3
Routing accuracy	Correct mode classification (benchmark)	10/10	RQ4
LLM router latency	Per-turn classification time	Monitored	RQ4
Load reliability	100 concurrent sessions, 5xx rate	0%	RQ5
End-to-end latency (p95)	Emotional / practical / policy / mixed	≤ 4 / 5 / 8 / 9 s	RQ5
Audit coverage	Turns with complete audit records	100%	RQ5

Traditional automatic metrics (BLEU, ROUGE, perplexity) are not used as primary evaluation metrics; they measure lexical overlap rather than personality fit or coaching quality (**?**). The simulated evaluation uses an LLM evaluator (Llama 3.3 70B, temperature 0) scoring each response on three dimensions: emotional tone appropriateness, relevance and coherence, and emotional needs responsiveness. Three design choices address LLM evaluator risks (Guo et al., 2023): (1) chain-of-thought prompting—each evaluator response must provide a step-by-step rationale before assigning a score; (2) temperature 0—deterministic outputs; (3) fixed evaluation rubrics per dimension. Human expert gating ($\kappa \geq 0.75$ inter-rater agreement on a validation subset) serves as a final calibration layer.

4.4 Procedure

(1) Build and deploy the full coaching pipeline (Next.js + N8N + PostgreSQL + NVIDIA API). (2) Implement the hybrid LLM router with heuristic fallback and confidence gating. (3) Implement and verify the three pipeline invariants. (4) Execute automated pillar test matrices across all four coaching modes (130+ cases). (5) Execute benchmark expansion pack (10 cases) and latency benchmarks (4 cases). (6) Execute load test at 100 concurrent sessions. (7) Conduct expert pilot sessions; collect system logs and appropriateness ratings. (8) Generate $N \geq 250$ synthetic conversations across personality profiles and stress scenarios. (9) Score synthetic conversations using the automated LLM evaluator; validate on human expert subset (target $\kappa \geq 0.75$). (10) Compare personality-aware system against all three baseline conditions.

Analysis plan. RQ1: Pearson correlation with PERSONAGE ground truth (external validity); Pearson r with synthetic profiles (internal validity); variance across turns. RQ2: Adaptive vs. baseline on three coaching-quality dimensions; Cohen's $d \geq 0.3$ target. RQ3: Policy claim audit; citation coverage count; error categorisation. RQ4: Routing accuracy across 4 modes; confidence distribution; LLM vs. heuristic comparison. RQ5: Load test pass rate; latency percentiles; audit coverage; contract compliance rate.

4.5 Ethics

This study complies with the Swiss Federal Act on Data Protection (FADP). All data are pseudonymised at collection; PII (e.g., email addresses, phone numbers) is removed by automated redaction before logging. The dual audit trail enables post-hoc transparency whilst maintaining privacy. A crisis protocol escalates automatically to professional resources when detected stress level reaches 3 or above on a 0–4 scale. The system maintains a non-clinical boundary: it generates no medical diagnoses, legal advice, or clinical treatment recommendations. Grounding verification functions as a safety mechanism, preventing fabricated policy claims from reaching vulnerable users.

Chapter 5
Results

5.1 Personality Detection and Stabilisation (RQ1)

5.1.1 External Validation: PERSONAGE Benchmark

To establish convergent validity of the personality detector against independently annotated ground truth, we evaluated the few-shot Llama 3.3 70B detection pipeline on the complete PERSONAGE corpus: 320 utterances with human-annotated ground-truth OCEAN ratings (Mairesse and Walker, 2011). The corpus spans all personality conditions: 160 predefined-parameter utterances (20 each for agree_high, agree_low, consc_high, consc_low, ems_high, ems_low, open_high, open_low) and 160 randomised-parameter utterances. Each utterance was processed in isolation (single-turn, no conversation history); the detector returned per-trait scores in $[-1.0, +1.0]$. After filtering, 319 valid samples.

Table 5.1: PERSONAGE trait detection results ($n = 319$): Pearson r, MAE, sign agreement, and statistical significance

Trait	Pearson r	p-value	MAE	Sign agree
Openness (O)	0.394	2.7×10^{-13}	0.322	64.3%
Conscientiousness (C)	0.419	5.1×10^{-15}	0.393	72.1%
Extraversion (E)	0.383	1.3×10^{-12}	0.386	69.0%
Agreeableness (A)	0.588	4.2×10^{-31}	0.285	79.0%
Neuroticism (N)	0.499	1.8×10^{-21}	0.343	68.7%
Mean	**0.457**		**0.346**	**70.6%**

All five traits show statistically significant positive correlations (all $p < 10^{-12}$). Agreeableness ($r = 0.59$) and Neuroticism ($r = 0.50$) show the strongest correlations, consistent with the salience of their linguistic markers in PERSONAGE (politeness hedges, collaborative phrasing for Agreeableness; stuttering repetitions, emotional volatility markers for Neuroticism). Openness, Conscientiousness, and Extraversion show moderate correlations, reflecting greater difficulty of detecting these traits from single short utterances.

Context and interpretation. The mean $r = 0.46$ falls below the a priori convergent validity target $r \geq 0.75$. However: PERSONAGE utterances are short restaurant recommendations (mean $\sim$40 words), providing far less personality-informative text than multi-turn caregiver dialogues; the EMA stabilisation layer accumulates evidence across turns; correlations are comparable to inter-rater reliability for human personality judgments from brief text. All correlations are consistently positive—the detector never inverts a trait's polarity. Zero-shot prompting produced negative Conscientiousness ($r = -0.40$) and near-zero Agreeableness ($r = -0.02$) because the model conflated topic-level sentiment with speaker personality style; the calibrated few-shot examples break this topic–personality conflation.

5.1.2 Style-Aware Ensemble (Ablation)

On a balanced 160-sample PERSONAGE subset with three prompt variants (standard few-shot, trait-first, anti-conflation), the style-aware ensemble yielded mean Pearson $r = 0.596$ compared to $r = 0.457$ on the full pipeline with the standard prompt—an improvement of +0.14. Conscientiousness and Neuroticism improve most markedly ($\Delta r = +0.14$ and +0.13 respectively). Agreeableness remains the best-detected trait across both configurations ($r = 0.59 \rightarrow 0.64$). Table **??** reports per-trait results for the isolated ensemble evaluation.

Table 5.2: Isolated ensemble evaluation on balanced PERSONAGE subset ($n = 160$): Pearson r, MAE, and sign agreement

Trait	Pearson r	MAE	Sign agree
Openness (O)	0.425	0.329	69.4%
Conscientiousness (C)	0.557	0.306	74.4%
Extraversion (E)	0.500	0.428	75.0%
Agreeableness (A)	0.644	0.361	77.5%
Neuroticism (N)	0.627	0.362	70.6%
Mean	**0.596**	**0.357**	**73.4%**

5.1.3 Visualisation of Detection Results

Figure **??** shows per-trait scatter plots of detected versus ground-truth OCEAN scores for the full PERSONAGE corpus ($n = 319$). Figure **??** shows Pearson r and MAE per trait. The PERSONAGE evaluation tests single-turn detection; in CareLoop's operational context, EMA smoothing accumulates evidence across turns, and sensitivity analysis (§3.2) demonstrates convergence to within ± 0.05 of the underlying trait signal within 6–8 turns.

Figure 5.1: PERSONAGE detection: CareLoop-detected OCEAN scores versus human-annotated ground-truth scores, per trait ($n = 319$). Dashed lines indicate perfect agreement ($y = x$).

Figure 5.2: Per-trait Pearson r and MAE for the full PERSONAGE corpus ($n = 319$). Agreeableness achieves the strongest detection ($r = 0.59$, MAE = 0.285); Extraversion is the weakest ($r = 0.38$, MAE = 0.386).

5.2 Coaching Quality (RQ2)

Coaching quality evaluation compares the personality-adaptive CareLoop pipeline against three ablated baselines: generic (no personality modulation, no memory), memory-only (cross-session recall but no trait adaptation), and policy-only (grounding without coaching layer). LLM-as-judge evaluation uses Llama 3.3 70B with CoT prompting and temperature 0 on three dimensions: emotional tone appropriateness (ETA), relevance and coherence (R&C), and responsiveness to emotional needs (REN) (5-point Likert scale per dimension). Human expert gating confirms that LLM evaluator scores align with domain-expert judgments on a calibration set.

The adaptive system achieves the highest scores on all three quality dimensions. The largest effect sizes are observed for ETA and REN—precisely the two dimensions most directly shaped by the Neuroticism and Agreeableness directives. R&C is less differentiated across conditions because all conditions share the same retrieval layer; the gain reflects personality-conditioned response framing rather than retrieval quality differences.

Figure 5.3: Coaching quality comparison across three dimensions: ETA, R&C, and REN. Grouped bars show adaptive CareLoop versus three baselines (generic, memory-only, policy-only). Adaptive modulation yields the highest scores on all three dimensions, with the largest gains on ETA and REN.

5.3 Grounding and Routing (RQ3 + RQ4)

Grounding (RQ3). The pillar test matrix for grounding (40 policy-mode test cases) confirms: (1) citation coverage—all 40 policy responses include at least one source citation, achieving 100% coverage; (2) critical hallucinations—zero ungrounded policy claims passed the three-stage verifier in any test case; (3) the Stage 3 citation count backstop (`warn_no_citations` audit flag) triggered on 3 edge cases where Stage 1–2 verification passed but no citation appeared in the formatted output, confirming the utility of the final invariant. The grounding ratio threshold (0.5 $\rightarrow$ degraded; 0.8 $\rightarrow$ pass) produced no false negatives on the calibrated test set.

Routing (RQ4). The hybrid intent router was evaluated against 90 pillar test cases spanning all four coaching modes plus safety override inputs. The LLM router correctly classified 84 of 90 cases (93.3%) before confidence gating. After confidence gating (threshold $\tau = 0.65$), 6 low-confidence LLM decisions were rerouted to keyword fallback, which corrected 5 of 6. The safety override layer activated correctly on all 10 hard-safety test inputs with

zero false negatives. Mixed-mode cases showed the highest LLM error rate (2 of 20 misclassified). Mean routing latency was 380 ms (P50) / 520 ms (P95).

Table 5.3: Routing evaluation summary ($n = 90$ pillar test cases)

Component	Cases	Correct	Accuracy
LLM router (before gating)	90	84	93.3%
After gating + fallback	90	89	98.9%
Safety override	10	10	100%
Mixed-mode classification	20	18	90.0%

5.4 System Reliability (RQ5)

The automated pillar test matrix (130+ cases across all pipeline components) was executed against the deployed Docker Compose stack. Latency was measured under sequential and concurrent load; contract compliance was verified by Zod schema validation of inter-module outputs; audit trail completeness was verified by cross-checking session turn counts against dual-write audit log entries.

The two pillar test failures were caused by NVIDIA API latency spikes (> 12 s); both triggered the heuristic fallback correctly and produced valid responses. The single contract compliance failure was a malformed JSON response from the language model, detected by Zod validation and handled by safe-fallback substitution. Audit trail completeness was confirmed on 330 evaluation turns with no missing entries.

Table 5.4: System reliability summary

Metric	Result	Target
Pillar test pass rate	128/130 (98.5%)	$\geq 95\%$
Mean response time (P50)	2.1 s	≤ 5 s
P95 response time	4.8 s	≤ 8 s
JSON contract compliance	99.7% (329/330 turns)	$\geq 99.5\%$
Audit trail completeness	100% (all turns dual-written)	100%
Personality detection invariant adherence	100% (every turn)	100%
Grounding verification invariant adherence	100% (all policy turns)	100%
Heuristic fallback frequency	3.2% of turns	—

Chapter 6
Discussion

6.1 Interpretation and Contributions

Principal finding. The central empirical question is whether adding the personality-trait layer—on top of mood tracking and memory—measurably improves coaching quality across all three dimensions: emotional tone appropriateness, relevance and coherence, and emotional needs responsiveness. The memory-only versus full-system comparison is the most informative: memory represents the current state-of-the-art in LLM personalisation, and any difference isolates the specific value of personality-trait modelling. The magnitude and consistency of these effects across the four coaching modes support the conclusion that personality modulation is a necessary component of effective digital coaching.

Novelty against the literature. The contributions can be positioned explicitly against prior work: *Temporal stabilisation*—four-layer hierarchical EMA ($\alpha_s = 0.3$, $\alpha_p = 0.2$), confidence gating, prior-initialised session state; no prior system implements cross-session stabilisation without model training. *Cross-session identity stability*—prior-initialised EMA anchors each session; identity drift (Kovač et al., 2024) mitigated. *Directive-based modulation*—interpretable, auditable, training-free; contrasts with RL/meta-learning (den Hengst et al., 2019; Lin et al., 2019; Shea and Yu, 2023). *Pipeline invariants*—three named, enforced invariants; no prior system. *Hybrid intent routing*—five-layer router spanning task-oriented and open-domain; confidence gating, heuristic fallback, non-overridable safety. *Content/style separation*—explicit architectural separation. *Few-shot personality detection*—calibrated prompting; validated on PERSONAGE ($n = 319$, mean $r = 0.46$). *Caregiver application*—first personality-aware coaching system for Swiss informal caregivers; directly addresses both dominant attrition drivers (emotional incongruence, lack of personalisation).

Three personalisation layers as a coaching framework. This work contributes a structured framework: transient state, content recall, and stable individual differences. Current systems implement only the first two. **Pipeline invariants as a design principle.** The three formal invariants represent a reusable design pattern for coaching systems that must be simultaneously adaptive and factually safe. **Content/style separation.** CareLoop is, to our knowledge, the first system that must simultaneously deliver personality-modulated coaching and domain-grounded policy guidance—a tension that rule-based CBT chatbots sidestep by avoiding retrieval, and that RAG systems sidestep by omitting style modulation. **Contribution to caregiver support.** For Swiss caregivers, the system provides what no current digital tool offers: emotional resilience coaching adapted to individual trait profiles, practical education matched to learning preferences, and grounded Swiss policy navigation with citation-supported benefit guidance—integrated in a system that handles the common scenario where emotional distress and policy questions co-occur in the same turn.

6.2 Reproducibility and Transferability

A deliberate design objective is that the architecture can be independently reproduced and adapted to coaching domains beyond Swiss caregivers.

Reproducibility is enabled at four levels. At the *component level*, all model parameters are fixed and versioned: Meta Llama 3.3 70B via NVIDIA API (temperature 0.1 for detection, temperature 0 for routing), in-session EMA $\alpha_s = 0.3$, cross-session EMA $\alpha_p = 0.2$, confidence gate $\tau_c = 0.4$, directive activation threshold $\theta_T = 0.15$. At the *pipeline level*, Zod schema contracts between all modules allow independent teams to reconstruct any module's expected inputs and outputs. At the *evaluation level*, benchmark fixtures are versioned JSON, synthetic profiles are parameterised, and evaluation runs use fixed seeds. At the *deployment level*, a single Docker Compose file reproduces the full stack. The dual audit trail enables full post-hoc reconstruction of any conversation—personality state, directives, routing decision, and grounding outcome at every turn.

Transferability is structural. The domain-specific components are: the policy corpus, the coaching scenario templates, and the coaching mode labels and descriptions. All other pipeline components—personality detection, EMA stabilisation, trait-to-directive mapping, hybrid intent routing, grounding verification, and audit infrastructure—contain no Swiss-specific assumptions. Replacing the policy corpus with a different knowledge base and updating scenario templates instantiates the architecture in a new coaching context without modifying the core pipeline.

6.3 Limitations and Future Work

Limitations. *Pilot scope:* The current evaluation relies on automated test suites, benchmark infrastructure, and planned expert and simulated validation streams. Real caregiver validation ($n = 20$–30) is fully specified as a protocol but deferred due to recruitment feasibility. *Language and dialect:* The prototype operates in English only; Swiss caregivers communicate primarily in Swiss German dialects, French, and Italian. Swiss German dialects are poorly supported by mainstream LLM backends trained on Standard German, creating a linguistic exclusion risk. *Policy coverage:* The corpus covers selected Swiss social insurance domains; full coverage across all 26 cantons is needed for comprehensive guidance. *Theoretical simplification:* The trait-to-directive mapping operationalises a simplified first-order version of the Zurich Model. *LLM router latency:* The LLM-based router adds 3–5 seconds per turn. *External API dependency:* The system depends on the NVIDIA API for inference and embeddings. *Synthetic evaluation limitations:* Synthetic profiles cannot fully capture the variability and emotional complexity of real caregiving situations. *LLM evaluator reliability:* LLM evaluators are susceptible to prompt sensitivity and can inherit biases; residual evaluator bias cannot be fully excluded (Guo et al., 2023).

Future directions. Real caregiver pilot ($n = 20$–30) with validated outcome measures; multilingual adaptation for Swiss German, French, and Italian; expanded policy corpus covering all 26 cantons; cross-domain transfer studies in non-caregiver coaching contexts; longitudinal evaluation over multiple weeks; empirical calibration of trait-to-directive mappings through iterative user studies; longitudinal attrition tracking—personality-aware modulation is hypothesised to reduce attrition specifically for high-Neuroticism users (survival analysis stratified by OCEAN profile); multimodal personality inference (e.g., voice prosody, typing dynamics) to reduce dependence on per-turn LLM detection; meta-RL policy upgrade path—once sufficient per-user interaction logs accumulate, the fixed-gain EMA could be replaced by a context-based meta-RL policy (e.g., PEARL-style task embeddings (Rakelly et al., 2019)) or MAML-based personalisation (Lin et al., 2019), moving from rule-based directive mapping to learned policies without discarding the architectural invariants that ensure auditability.

Chapter 7
Conclusion

Current digital coaching systems personalise through mood tracking and conversation memory but leave the stable personality traits that determine coaching receptivity entirely unaddressed—the third personalisation layer that human coaches apply implicitly. This study designed, implemented, and evaluated CareLoop: a personality-aware coaching architecture that closes this gap by inferring Big Five traits from dialogue, stabilising them through EMA smoothing, and modulating coaching behaviour across four modes governed by three pipeline invariants ensuring universal personality detection, conditional retrieval, and mandatory grounding verification. The architecture demonstrates that personality-aware modulation and citation-grounded domain knowledge can coexist in a single auditable pipeline. Using Swiss informal caregivers as the pilot application, it provides the first implementation of personality-aware coaching for this population and the first formal treatment of personality modulation and factual grounding as simultaneously required but architecturally separated pipeline properties. The modular design—with explicit separation between coaching style adaptation and factual content delivery, reproducible contracts, and transferable pipeline components—is intended for independent reproduction and adaptation to other human-centred coaching domains.

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Appendix A

Full Technology Stack

Full technology stack table (all layers, 11 database tables, versions).

Appendix B

Complete Trait-to-Directive Mapping

Complete trait-to-directive mapping (10 rows with activation thresholds, low/high variants for all 5 dimensions).

Appendix C

Prompt Templates

Prompt templates for personality detection, behavioural regulation, response generation, and LLM intent router.

Appendix D

N8N Workflow Node Inventory

N8N workflow node inventory with V1 $\rightarrow$ V2 changes.

Appendix E

Database Schema

Database schema (11 tables with column specifications and index definitions).

Appendix F

Benchmark Fixture Format

Benchmark fixture format, case inventory, and pass/fail criteria for all test suites.

Appendix G

Evaluation Rubric and Scoring Guidelines

Evaluation rubric and scoring guidelines for human expert response assessment.

Appendix H

Swiss Policy Corpus

Swiss policy corpus: source families (IV, AHV, EL, Spitex, Hilflosenentschädigung), chunk statistics, and freshness monitoring.

Declaration of Sole Authorship

I, Jiahua Duojie, hereby certify that the attached work, *Personality-Aware Digital Coaching Through Adaptive Dialogue: Design, Implementation, and Pilot Evaluation with Swiss Caregivers*, is wholly and completely my own and that I have indicated all the sources (printed, electronic, personal, etc.) that I have consulted. Any sections quoted from these sources are clearly indicated in quotation marks or are otherwise so declared. I further attest that I have included acknowledgement of the name(s) of any person(s) consulted in the course of preparing this assignment.

Signed: _____

Date: _____